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TITLE: Sequence-to-Sequence Tasks — Using Hugging Face Pipelines for Translation

STEP 1: Install and Import Required Libraries

```
!pip install transformers
!pip install torch
!pip install sentencepiece

from transformers import pipeline
print("Transformers installed successfully!")
```

```
Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-pac
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/d
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/
Requirement already satisfied: typer-slim in /usr/local/lib/python3.12/dist-p
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.1
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-p
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.12/
Requirement already satisfied: hf-xet<2.0.0,>=1.3.2 in /usr/local/lib/python3
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packag
Requirement already satisfied: typing-extensions>=4.1.0 in /usr/local/lib/pyt
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packag
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-pack
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dis
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-package
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-pa
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.1
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/pyth
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-p
Requirement already satisfied: torch in /usr/local/lib/python3.12/dist-packag
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-pac
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/py
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-p
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dis
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/d
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packa
```

```
Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dis
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.1
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/d
Requirement already satisfied: sentencepiece in /usr/local/lib/python3.12/dis
Transformers installed successfully!
```

STEP 2: Load Translation Model

```
from transformers import AutoTokenizer, AutoModelForSeq2SeqLM

# For English to French
tokenizer_en_fr = AutoTokenizer.from_pretrained("Helsinki-NLP/opus-mt-en-fr"
model_en_fr = AutoModelForSeq2SeqLM.from_pretrained("Helsinki-NLP/opus-mt-en

# For English to German
tokenizer_en_de = AutoTokenizer.from_pretrained("Helsinki-NLP/opus-mt-en-de"
model_en_de = AutoModelForSeq2SeqLM.from_pretrained("Helsinki-NLP/opus-mt-en

# For English to Hindi
tokenizer_en_hi = AutoTokenizer.from_pretrained("Helsinki-NLP/opus-mt-en-hi"
model_en_hi = AutoModelForSeq2SeqLM.from_pretrained("Helsinki-NLP/opus-mt-en

text = "Machine learning is transforming modern technology."

# Translate English to French
input_ids_en_fr = tokenizer_en_fr(text, return_tensors="pt").input_ids
generated_ids_en_fr = model_en_fr.generate(input_ids_en_fr)
translated_text_en_fr = tokenizer_en_fr.decode(generated_ids_en_fr[0], skip_
print([{'translation_text': translated_text_en_fr}])

# Translate English to German
input_ids_en_de = tokenizer_en_de(text, return_tensors="pt").input_ids
generated_ids_en_de = model_en_de.generate(input_ids_en_de)
translated_text_en_de = tokenizer_en_de.decode(generated_ids_en_de[0], skip_
print([{'translation_text': translated_text_en_de}])

# Translate English to Hindi
input_ids_en_hi = tokenizer_en_hi(text, return_tensors="pt").input_ids
generated_ids_en_hi = model_en_hi.generate(input_ids_en_hi)
translated_text_en_hi = tokenizer_en_hi.decode(generated_ids_en_hi[0], skip_
print([{'translation_text': translated_text_en_hi}])
```

```
/usr/local/lib/python3.12/dist-packages/transformers/models/marian/tokenizati
  warnings.warn("Recommended: pip install sacremoses.")
Loading weights: 100%  258/258 [00:00<00:00, 484.74it/s, Materializing param=model.shared.we

The tied weights mapping and config for this model specifies to tie model.sha
The tied weights mapping and config for this model specifies to tie model.sha
Loading weights: 100%  258/258 [00:00<00:00, 694.19it/s, Materializing param=model.shared.we

The tied weights mapping and config for this model specifies to tie model.sha
The tied weights mapping and config for this model specifies to tie model.sha
Loading weights: 100%  258/258 [00:00<00:00, 532.98it/s, Materializing param=model.shared.we

The tied weights mapping and config for this model specifies to tie model.sha
The tied weights mapping and config for this model specifies to tie model.sha
[{'translation_text': "L'apprentissage automatique transforme la technologie
[{'translation_text': 'Maschinelles Lernen transformiert moderne Technologie.
[{'translation_text': 'मशीन सीखने आधुनिक तकनीक परिवर्तित कर रहा है।'}]
```

STEP 3: Prepare Input Sentences

```
sentences = [

# Simple
"The weather is pleasant today.",
"Artificial intelligence is changing the world.",

# Compound
"I finished my assignment, and I submitted it online.",
"The sun was shining, but the wind was cold.",

# Complex
"Although the project was difficult, the team completed it successfully.",
"If you study regularly, you will achieve good results.",

# Interrogative
"What time does the meeting start?",
"Why is machine learning important in data science?",

# Technical
"Neural networks are widely used in deep learning applications.",
"Data preprocessing improves machine learning model performance.",

# Conversational
"Hey, how are you doing today?",
"Let's meet after class for coffee.",

# News
"The government announced new economic policies today.",

# Academic
"Research in natural language processing has advanced significantly.",

# Social media
```

```
    "This new smartphone camera is amazing!"
    ]
```

STEP 4: Perform Translation

```
print("English → French\n")

for sentence in sentences:
    input_ids = tokenizer_en_fr(sentence, return_tensors="pt").input_ids
    generated_ids = model_en_fr.generate(input_ids)
    translated_text = tokenizer_en_fr.decode(generated_ids[0], skip_special_
    print("English:", sentence)
    print("French:", translated_text)
    print()
```

```
English → French

English: The weather is pleasant today.
French: Le temps est agréable aujourd'hui.

English: Artificial intelligence is changing the world.
French: L'intelligence artificielle change le monde.

English: I finished my assignment, and I submitted it online.
French: J'ai terminé ma mission, et je l'ai soumise en ligne.

English: The sun was shining, but the wind was cold.
French: Le soleil brillait, mais le vent était froid.

English: Although the project was difficult, the team completed it successful
French: Bien que le projet ait été difficile, l'équipe l'a mené à bien.

English: If you study regularly, you will achieve good results.
French: Si vous étudiez régulièrement, vous obtiendrez de bons résultats.

English: What time does the meeting start?
French: A quelle heure commence la réunion ?

English: Why is machine learning important in data science?
French: Pourquoi l'apprentissage automatique est-il important en sciences des

English: Neural networks are widely used in deep learning applications.
French: Les réseaux neuraux sont largement utilisés dans les applications d'a

English: Data preprocessing improves machine learning model performance.
French: Le prétraitement des données améliore les performances des modèles d'

English: Hey, how are you doing today?
French: Comment ça va aujourd'hui ?

English: Let's meet after class for coffee.
French: Rendez-vous après le cours pour prendre un café.

English: The government announced new economic policies today.
French: Le gouvernement a annoncé aujourd'hui de nouvelles politiques économi

English: Research in natural language processing has advanced significantly.
```

```
French: La recherche dans le domaine du traitement des langues naturelles a c

English: This new smartphone camera is amazing!
French: Cette nouvelle caméra pour smartphone est incroyable !
```

STEP 5: Translate into Multiple Languages

```
for sentence in sentences:

    # Translate English to French
    input_ids_fr = tokenizer_en_fr(sentence, return_tensors="pt").input_ids
    generated_ids_fr = model_en_fr.generate(input_ids_fr)
    fr = tokenizer_en_fr.decode(generated_ids_fr[0], skip_special_tokens=Tru

    # Translate English to German
    input_ids_de = tokenizer_en_de(sentence, return_tensors="pt").input_ids
    generated_ids_de = model_en_de.generate(input_ids_de)
    de = tokenizer_en_de.decode(generated_ids_de[0], skip_special_tokens=Tru

    # Translate English to Hindi
    input_ids_hi = tokenizer_en_hi(sentence, return_tensors="pt").input_ids
    generated_ids_hi = model_en_hi.generate(input_ids_hi)
    hi = tokenizer_en_hi.decode(generated_ids_hi[0], skip_special_tokens=Tru

    print("English:", sentence)
    print("French :", fr)
    print("German :", de)
    print("Hindi  :", hi)
    print("-"*60)
```

```
German : Die Sonne schien, aber der Wind war kalt.
Hindi  : सूरज चमकने लगा था, लेकिन हवा ठंडा थी ।
------------------------------------------------------------
English: Although the project was difficult, the team completed it successfu
French : Bien que le projet ait été difficile, l'équipe l'a mené à bien.
German : Obwohl das Projekt schwierig war, hat das Team es erfolgreich abges
Hindi  : हालाँकि यह काम बहुत मुश्किल था, फिर भी टीम ने इसे सफलतापूर्वक पूरा किया ।
```

```
German : Neurale Netzwerke werden häufig in Deep-Learning-Anwendungen einges
Hindi  : तंत्रिका नेटवर्कों को गहरी सीखने की अनुप्रयोगों में व्यापक रूप से उपयोग किया जाता है
------------------------------------------------------------
English: Data preprocessing improves machine learning model performance.
French : Le prétraitement des données améliore les performances des modèles
German : Die Datenvorverarbeitung verbessert die Leistungsfähigkeit des Mode
Hindi  : डाटा पूर्व प्रक्रिया करने से मशीन सीखने का मॉडल प्रदर्शन बेहतर होता है.
------------------------------------------------------------
English: Hey, how are you doing today?
French : Comment ça va aujourd'hui ?
German : Hey, wie geht's dir heute?
Hindi  : अरे, आप आज कैसे कर रहे हैं?
------------------------------------------------------------
English: Let's meet after class for coffee.
French : Rendez-vous après le cours pour prendre un café.
German : Treffen wir uns nach dem Unterricht zum Kaffee.
Hindi  : चलो वर्ग के बाद कॉफी के लिए मिलते हैं.
------------------------------------------------------------
English: The government announced new economic policies today.
French : Le gouvernement a annoncé aujourd'hui de nouvelles politiques econo
German : Die Regierung kündigte heute eine neue Wirtschaftspolitik an.
Hindi  : सरकार ने आज नई आर्थिक नीतिओं की घोषणा की.
------------------------------------------------------------
English: Research in natural language processing has advanced significantly
French : La recherche dans le domaine du traitement des langues naturelles a
German : Die Forschung in der natürlichen Sprachverarbeitung hat sich deutli
Hindi  : कुदरती भाषा में खोजबीन करने से और भी कई फायदे हुए हैं ।
------------------------------------------------------------
English: This new smartphone camera is amazing!
French : Cette nouvelle caméra pour smartphone est incroyable !
German : Diese neue Smartphone-Kamera ist erstaunlich!
Hindi  : यह नया स्मार्टफोन कैमरा कमाल है!
------------------------------------------------------------
```

STEP 6: Translation Quality Analysis

The translation results show that pretrained transformer models produce highly fluent and grammatically correct translations for most simple sentences. Short and straightforward sentences are translated with high accuracy across all target languages. However, complex sentences sometimes show slight structural changes, especially in German and Hindi translations. Technical terms such as "neural networks" and "machine learning" are usually preserved or translated accurately due to the model's exposure to technical datasets. Conversational sentences sometimes appear slightly more formal in translation compared to the original English text. The model generally maintains the core meaning of sentences even when minor word order changes occur. Some idiomatic expressions may not translate perfectly due to cultural differences between languages. Overall, the translation quality is good for general-purpose text but may require human correction for specialized domains.

STEP 7: Testing Complex Sentences

```
complex_sentences = [
"Although artificial intelligence has many benefits, it also raises ethical
"If researchers continue improving natural language models, machines may soo
"Can machine learning algorithms truly understand human emotions?",
"The research paper explains how transformer architectures improved translat
"Even though the software was difficult to install, the students successfull
]

for sentence in complex_sentences:
    input_ids = tokenizer_en_fr(sentence, return_tensors="pt").input_ids
    generated_ids = model_en_fr.generate(input_ids)
    fr = tokenizer_en_fr.decode(generated_ids[0], skip_special_tokens=True)

    print("English:", sentence)
    print("French:", fr)
    print()
```

```
English: Although artificial intelligence has many benefits, it also raises e
French: Bien que l'intelligence artificielle ait de nombreux avantages, elle

English: If researchers continue improving natural language models, machines
French: Si les chercheurs continuent d'améliorer les modèles de langage natur

English: Can machine learning algorithms truly understand human emotions?
French: Les algorithmes d'apprentissage automatique peuvent-ils vraiment comp

English: The research paper explains how transformer architectures improved t
French: Le document de recherche explique comment l'architecture des transfor

English: Even though the software was difficult to install, the students succ
French: Même si le logiciel était difficile à installer, les étudiants ont ré
```

STEP 8: Lab Report

Dataset Description

The dataset used in this experiment consists of 15 English sentences collected from different sources such as news articles, academic texts, daily conversations, and social media posts. The purpose of selecting sentences from multiple domains is to evaluate how the translation model performs across different writing styles and contexts. The dataset includes simple, compound, complex, interrogative, technical, and conversational sentences. This diversity allows better observation of the model's ability to handle different linguistic structures. The sentences were manually selected to represent real-world language usage.

Conclusion

This experiment demonstrated the use of Seq2Seq Transformer models for machine translation using the Hugging Face Transformers library. Pretrained translation pipelines allow users to perform multilingual translation without training a model from

scratch. The results show that Transformer-based models produce accurate and fluent translations for most sentences. However, complex sentence structures and idiomatic expressions can still pose challenges for automatic translation systems. Overall, pretrained models provide an efficient and practical solution for real-world language translation tasks.